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SUPERSEDING
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MILITARY SPECIFICATION

BEARING, ROLLER, THRUST

This specification was approved by the Departments of the Army, the Navy and the Air Force.

1. SCOPE

1.1 This specification covers thrust bearings containing cylindrical rollers for use where loads are parallel to the shaft axis.

2. APPLICABLE SPECIFICATIONS, STANDARDS, DRAWINGS, AND PUBLICATIONS

2.1 The following specifications, standards, drawings, and publications, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL

QQ-M-151 —Metals; General Specification for Inspection of

MILITARY

MIL-P-197 — Preservation, Packaging, and Packing of Antifriction Bearings and Bearing Parts.

NAVY DEPARTMENT

Appendix II, Metals; Part A of General Specifications for Inspection of Material.

STANDARDS

MILITARY

MIL-STD-105— Sampling Procedures and Tables for Inspection by Attributes.

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 **Construction.** — Roller bearings shall consist of two flat thrust plates (or races), cylindrical rollers, and a separator (see figure 1). The rollers shall be of the straight, solid, type, of different lengths, with the long and short rollers alternately arranged in the slots to equalize wear. The ends of the rollers may be machined to form a radius. The rollers shall be retained in the separator when the bearing is disassembled. The slots in the separator shall admit the lubricant freely to the rollers and races.

3.2 **Surface condition.**—The surfaces listed in table I shall be ground, to the maximum surface roughness indicated. Ground surfaces shall be free from tool marks, chatter waves, pits, rust, or other injurious defects. Other surfaces shall have a smooth machined finish free from scale.

TABLE I.—*Surface roughness*

Surface	Maximum surface roughness number
Bearing surface of the rollers	15 RMS
Flat faces of the races	15 RMS
Outside diameter of the race piloted in the housing	30 RMS
Inside diameter of race piloted in the shaft	30 RMS

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3.3 **Material.**

3.3.1 *Races and rollers.*—Unless otherwise specified, the steel used for the manufacture of the rollers and races may be any high grade alloy steel which will produce finished articles that comply with the requirements of this specification. The metal shall be homogeneous in structure (except that case hardened material will be acceptable), free from pipes, seams, laminations, excessive inclusion of non-metallic impurities, and other injurious surface and internal defects. When specified in the contract or order, races and rollers shall be inspected for defects by magnetic particle methods. (See 6.1).

3.3.1.1 *Hardness of races and rollers.*—The hardness of the races and rollers shall be 58 to 63 on the Rockwell C scale.

3.3.1.2 *Drop test (races).*—The races shall not split, crack, chip, or be permanently deformed when dropped on edge a vertical distance of 4 feet on a mild steel plate, as specified in 4.3.3.

3.3.2 *Separator.*

3.3.2.1 *Material.*—Unless otherwise specified, the separator shall be wrought or cast bronze free from injurious defects. At the option of the bearing manufacturer, the separator may be provided with a steel insert or band to resist the outward thrust of the rollers.

3.3.2.1.1 *High speed applications.*—When specified, for high speed applications, the separator shall be provided with a hardened steel insert or band to resist the outward thrust of the rollers.

3.4 **Dimensions and tolerances.**

3.4.1 *Inside diameters, outside diameters, width.*—The inside diameters, outside diameters, width, and tolerances shall be as specified.

3.4.2 *Maximum shaft and housing fillet radii.*—The bearings shall clear the maximum shaft and housing fillet radii specified.

3.4.3 *Parallelism of races.*—Unless otherwise specified, the faces of the individual races shall be parallel within 0.001 inch.

3.4.4 *Rollers.*—The maximum variation in the diameter of any roller when measured at different positions shall be 0.00025 inch. The difference in maximum diameter between the largest roller and the smallest roller in any assembled set shall not exceed 0.00025 inch.

3.5 **Maximum working load and speed.**—The bearings shall operate satisfactorily at the maximum working load and speed specified (see 6.1).

3.6 **Markings.** — The bearings shall be marked with the manufacturer's name or symbol, and the manufacturer's part number.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 **Sampling**

4.1.1 *Lot size.* — Inspection lots shall be formed in accordance with MIL–STD–105.

4.1.2 *Dimensional and visual inspection.*—Sampling for dimensional and visual inspection (non-destructive) shall be in accordance with MIL–STD–105. The classification of defects and acceptable quality level shall be provided by the procuring activity.

4.1.3 *Tests, and inspection requiring disassembly.*—A sample of five bearings shall be subjected to tests, and such inspection as requires disassembly of the bearing. The number of samples shall be in addition to delivery requirements, and shall be furnished without cost to the Government. The lot shall be subject to rejection if any sample bearing fails to comply with requirements.

4.2 **Inspection.**

4.2.1 *General.*—Bearings shall be inspected

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by the Government inspector for requirements of this specification.

4.2.2 *Dimensional measurements.*

4.2.2.1 *Standards.* — The standards for measurements shall be gage blocks, plug, ring, or other special gages calibrated by the National Bureau of Standards at a temperature of 68° F. or checked as directed by the activity concerned.

4.2.2.2 *Temperature.* — All measurements shall be made after the standards, instruments, and the parts to be measured have been brought to the temperature of the room in which the measurements are to be made and which shall be 68° F. except as hereinafter provided. During the measurement of bearings or parts, and for a sufficiently stabilizing period preceding such measurements, neither the parts to be measured nor the measuring devices shall be subjected to direct sunlight, cold drafts, or other avoidable causes of change of temperature of a magnitude which may materially affect the accuracy of the measurements. Measurements may be made at a room temperature other than 68° F. provided this temperature is constant or slowly changing and further provided the standards are used to check the accuracy of the measurements; the standards being equally exposed to the same temperature conditions as the instruments and the parts to be measured. Measurements shall be corrected to 68° F. unless the standards have the same thermal coefficient of expansion as the parts measured.

4.2.2.3 *Measuring instruments.* — All dimensions shall be measured with suitable apparatus which will permit duplication of measurements to within one-half the last decimal unit of the dimension given.

4.3 Tests.

4.3.1 *General testing procedures.* — For Navy purchases, the general testing procedures shall be in accordance with Appendix II, Metals: Part A — Definitions and Tests. For purchases by other activities, general inspection procedures shall be in conformance with specification QQ–M–151.

4.3.2 *Hardness of races and rollers.*—Hardness tests shall be made at several points on the sample rollers and races separately, using the Rockwell C scale, 120 degree cone and 150 kilogram load, except that hardness tests on bearing surfaces shall be made with the Shore scleroscope.

4.3.3 *Drop test of races.*—The sample races shall be dropped on edge a vertical distance of 4 feet onto a mild steel plate approximately 1½ inches thick by 15 inches wide by 15 inches long.

5. PREPARATION FOR DELIVERY

5.1 Thrust bearings shall be cleaned, preserved, packaged, packed and marked in accordance with Specification MIL–P–197. The nomenclature shall be as follows: "Bearing, Roller, Thrust."

6. NOTES

6.1 **Ordering data.**—Invitations for bids and contracts or orders should specify the following:

a. Title, number and date of this specification.
b. Inside diameters and tolerances (see 3.4.1).
c. Outside diameters and tolerances (see 3.4.1).
d. Width of bearing, parallel to the axis, and tolerances (see 3.4.1).
e. Maximum shaft and housing fillet radii (see 3.4.2).
f. Maximum working load (see 3.5).
g. Operating speed (see 3.5).
h. Any desired options (see 3.3.1, 3.3.2.1.1).
i. Packaging options in MIL–P–197, including the following:

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(1) Whether bearings are for stock or production

(2) If for stock, the desired options in MIL–P–197.

(3) If for production, a method in accordance with MIL–P–197, if other than commercial practice is specified.

6.2 General construction. — The general construction of bearings covered by this specification is indicated in figure 1. This figure is illustrative only, and does not restrict the method used in manufacturing bearings.

Notice.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

Custodians:
Army—Ordnance Corps
Navy—Bureau of Ordnance
Air Force

Other interest:
Navy—Sh.

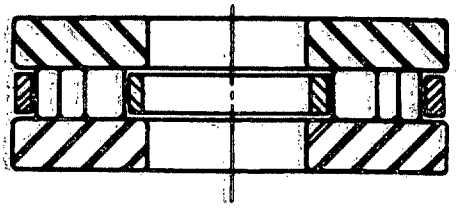


FIGURE 1.—*Construction.*

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